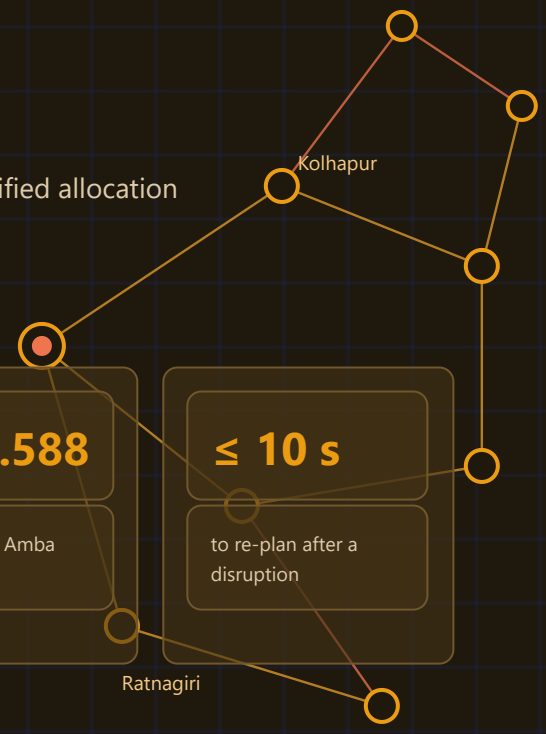
TEAM CYBERPUNK · TRIWIZATHON 2026 · PROBLEM STATEMENT 2

ARIA

Adaptive Routing and Inventory Allocation
Product Requirements Document

> Distribution Intelligence & Inventory Optimization Engine.
> Freight sets the service level — a calibrated-forecast, lot-aware, verified allocation engine for a Kolhapur super-stockist.



92%	₹11.03	0.856→0.588	≤ 10 s
of Indian grocery runs through kiranas	per t-km on a 6-ton truck vs ₹1.51 on 55-ton	τ* on Ratnagiri when Amba Ghat closes	to re-plan after a disruption

Round 1 PRD · v2.2 (final) · 24 October 2026
Sanjay Ghodawat University, Kolhapur

00 How to read this document

- **§1** is the whole story on one page.
- **§6.1** is the one idea that matters: *freight sets the service level*.
- **§4 J4** is the live demo.
- **§9** holds the pre-registered tests that decide whether ARIA works.
- The **appendices** hold the detail: validation rules, verifier codes, the greedy rule and the parameter registry.

Glossary. **τ** demand quantile level (share of days demand is at or below a quantity) · **τ*** the service level up to which a unit is worth shipping · **V*** best expected net value of a plan · **FEFO** first-expiry-first-out · **(a), b′, b″, (c)** the baselines of FR-P0-07.

Evidence labels

`FACT Rn` sourced.

`DESIGN` our decision.

`ASSUMPTION An` unverified, §10.2.

`MEASURED Mn` from our feasibility spike on synthetic data — timings and traps only.

Contents

01	Executive summary	the one-page story	02	Problem statement	evidence & pain
03	User personas	U1 · U2 · U3	04	Core user journeys	J1–J4 + demo
05	Functional requirements	P0 / P1 / P2	06	Innovation & differentiation	the one idea
07	Architecture & flow	diagrams, LP model, API	08	Feasibility & build plan	18-hour Gantt
09	Success metrics	KPI-1 … KPI-10	10	Data sources & assumptions	M1–M3, A1–A9
11	Limitations	stated plainly	A–D	Appendices & references	rules, verifier, registry

The system at a glance



FIG 0 · *ARIA in five verbs, and the practice it replaces.*

01 Executive Summary

Problem. About 92% of India's grocery retail runs through kiranas FACT R3 (figure unattributed in the article). Kiranas are fed by distributors who decide each evening which godown ships which product to which town. That decision has to handle three things at once:

Demand is a range

Tomorrow's sales are uncertain.

Freight varies

₹11.03 per t-km on a 6-ton vehicle vs ₹3.78 road average and ₹1.51 on a 55-ton trailer R1

Stock expires

A lot can be "in date" at the godown but too close to expiry when it reaches the store.

A rule such as "ship the forecast plus a fixed buffer from the nearest godown" A7 ignores all three.

User. U1 is a super-stockist with godowns at **Shiroli, Gokul Shirgaon and Kagal MIDC**, serving kirana beats in Kolhapur, Ichalkaranji, Jaysingpur, Kagal, Gadhinglaj and Ratnagiri (§3).

Solution. ARIA is a web app backed by a forecasting model and an optimizer. Each planning day it:

- **forecasts** demand per (town, product) as **calibrated quantiles** (LightGBM + conformal calibration);
- gives every godown → town route its own **freight-aware service level**: a unit ships only while its expected margin saved, plus the expected write-off it avoids, exceeds its freight and holding cost;
- allocates **lot by lot**, never shipping stock that would arrive below the store's minimum remaining shelf life, using shorter-life lots first;
- **verifies** every plan independently, gives every shortfall exactly one computed reason, and **re-routes** under disruptions such as the August 2026 Amba Ghat heavy-vehicle suspension R8.

"Routing" here means choosing each unit's godown → town route, not sequencing truck stops.

> THE ONE IDEA
>
> Freight per unit enters the service-level decision itself (§6.1). For an illustrative 1 kg, ₹100 product on a ~130 km Ratnagiri leg at a 3.5% margin, ARIA's stocking level is **0.856**. When heavy vehicles are banned and the leg must use 6-ton trucks it drops to **0.588**. A fixed safety factor (z = 1.65, or 0.95) ignores both facts.

Proof. ARIA runs on a distributor's own sales export (CSV import, FR-P0-01). We found no open Indian daily kirana/FMCG sales dataset (searched data.gov.in, Kaggle, Hugging Face, 24 Oct 2026; R27). So the engine is proven on real retail demand from the M5 benchmark R10: one M5 store's demand shape per town, rescaled to Census 2011 population R16, with NCAER's Indian freight costs R1.

A **fair replay** charges every policy one total-cost bill (freight + holding + write-off + lost margin) from the same starting stock on the same realised demand. Our spike showed why: without full-cost accounting, a naive policy "cut lost units by 86%" only by holding **5.6× more stock** M3.

12
P0 requirements, from data & import through Disruption Lab

2 LPs
value-max then exact FEFO tie-break; integral by total unimodularity

V1–V8
independent verifier; a failing plan is withheld

KPI-1
upper 95% CI of (ARIA − b") < 0 on total ₹ bill

Item	Value
AI components	ML: calibrated quantile forecasting (FR-P0-03). Planning: optimization over a network-flow model (FR-P0-06). Search and optimization are core AI methods R22
P0 scope	Data & CSV import; validation; forecasts; eligibility; freight-aware values; optimizer; baselines; verifier & metrics; fair replay; reason codes; Disruption Lab; API & UI

Item	Value
Headline KPI (§9)	**KPI-1:** total ₹ bill on realised demand, ARIA vs baseline **b″** (same optimizer and forecasts, one global service level). **Pass:** upper 95% CI bound of (ARIA − b″) < 0 — this tests the one idea directly
Demo	A judge applies "Amba Ghat closed to heavy vehicles". ARIA re-routes in ≤ 10 s next to the rule-based baseline and shows why Ratnagiri's service level fell (J4)

We claim no real-world savings. Results are method evidence on benchmark demand (§11).

02 Problem Statement

2.1 Official context

The brief asks for an engine that will "analyse available inventory, estimated demand, and distribution constraints to recommend which products should be sent to which locations, in what quantities, and from which warehouses". It should minimise "stockouts, unnecessary transportation costs, and inventory wastage", and "demonstrate how its decisions can improve supply-chain efficiency". Simulated data is allowed with stated assumptions `official statement`.

2.2 Evidence

Claim	Number	Source
Kirana share of Indian grocery retail (CY24)	**~92%**	R3 (stated by the article, unattributed)
India's logistics cost, FY2023-24	**7.97% of GDP**	R1; R2 secondary
Road freight: 55-ton / road average / 6-ton	**₹1.51 / ₹3.78 / ₹11.03** per t-km	R1 pp. 21, 83, 88
Warehousing	~₹30 per sq ft per month	R1 p. 118
Retail out-of-stock rate (multi-country, 2002)	8.3%	R4, not India-specific
Sales lost to stockouts	~4%	R5, secondary
FMCG distributor margin	~3.5–5%	R6; sources conflict
Near-expiry stock pushed back to distributors	AICPDF complaint	R7
Disruption on a Kolhapur route	Amba Ghat (NH-166) closed to heavy vehicles, Aug 2026	R8

Freight is a 7× swing — by vehicle class alone



FIG 1 · *Same road, same tonne — 7.3× the cost depending on the truck the lane forces you into.*

2.3 The decision that goes wrong

The chain runs **company → CFA depot → super-stockist → salesman beat → kirana**. Each evening U1 holds fixed stock across 3 godowns, in lots of different remaining life, against uncertain next-day demand in 6 towns.

Pain point	Rule-based practice (baseline a) A7	Consequence
Uncertain demand	Point forecast + fixed buffer (z = 1.65), nearest godown first	The same buffer everywhere, whatever the leg costs
Freight vs value	Ships on any open lane	On a 3.5% margin, a 6-ton Ratnagiri leg can cost more than the sale it protects
Shelf life vs transit	Expiry checked at the godown	Rejected deliveries; near-expiry stock stranded
Scarcity	First processed, first served	High-value demand short while low-value demand is filled
Disruption	Phone calls and manual edits	Nobody can say who loses and why
Proof	"Stockouts down X%"	Can be bought by over-stocking (M3)

2.4 Impact (illustrative, not a result)

₹4 L
sales lost per ₹1 crore turnover at ~4% stockouts [R5]

₹14–20k
margin lost per ₹1 crore at 3.5–5% margin [R6]

+₹7.25
extra per tonne-km on a 6-ton vehicle vs road average [R1]

These are retail averages for sell-out demand, not Kolhapur order data (A9). The replay replaces this sizing with measured, relative results.

03 User Personas

All personas are **composites** built from the cited evidence. **No interviews yet** (A1). A 10-minute on-site conversation with a Kolhapur distributor or kirana owner is planned for H0–1; its notes replace "assumed" wherever they differ.

U1 · "Prakash"

Super-stockist, family firm (owner-planner)

Goals. Plan ready before trucks load at 6 am; fewer stockout calls; lower freight per unit.

Pain. 6-ton legs at ₹11.03/t-km against a 3.5–5% margin; near-expiry pull-backs.

Constraints. Windows laptop; intermittent 4G; Marathi/English; no enterprise budget.

`FR-P0-05` `-06` `-12`

U2 · "Ravi"

Godown dispatch planner, Shiroli

Goals. Never dispatch stock that arrives short-dated; re-route within 10 s after a disruption.

Pain. Godown-level FEFO ignores transit time.

Constraints. Shared desktop; printable pick list.

`FR-P0-04` `-10` `-11`

U3 · "Aisha"

Supply analyst / brand planner

Goals. Compare policies in ₹ on identical data.

Pain. Each tool computes metrics differently.

Constraints. No time for model internals.

`FR-P0-07` `-08` `-09`

Scenarios (assumed, A1)

- **U1:** at 8 pm Kagal is short, Shiroli has surplus and the Ratnagiri leg is restricted. By 6 am Prakash needs to know which product goes where, from which godown.
- **U2:** at 7 am a truck is pulled off Amba Ghat. Ravi needs a new pick list before loading ends.
- **Stakeholder:** a kirana owner rejects a short-dated delivery and the units come back as a write-off. That cost is in ARIA's bill.

The distribution network ARIA plans over

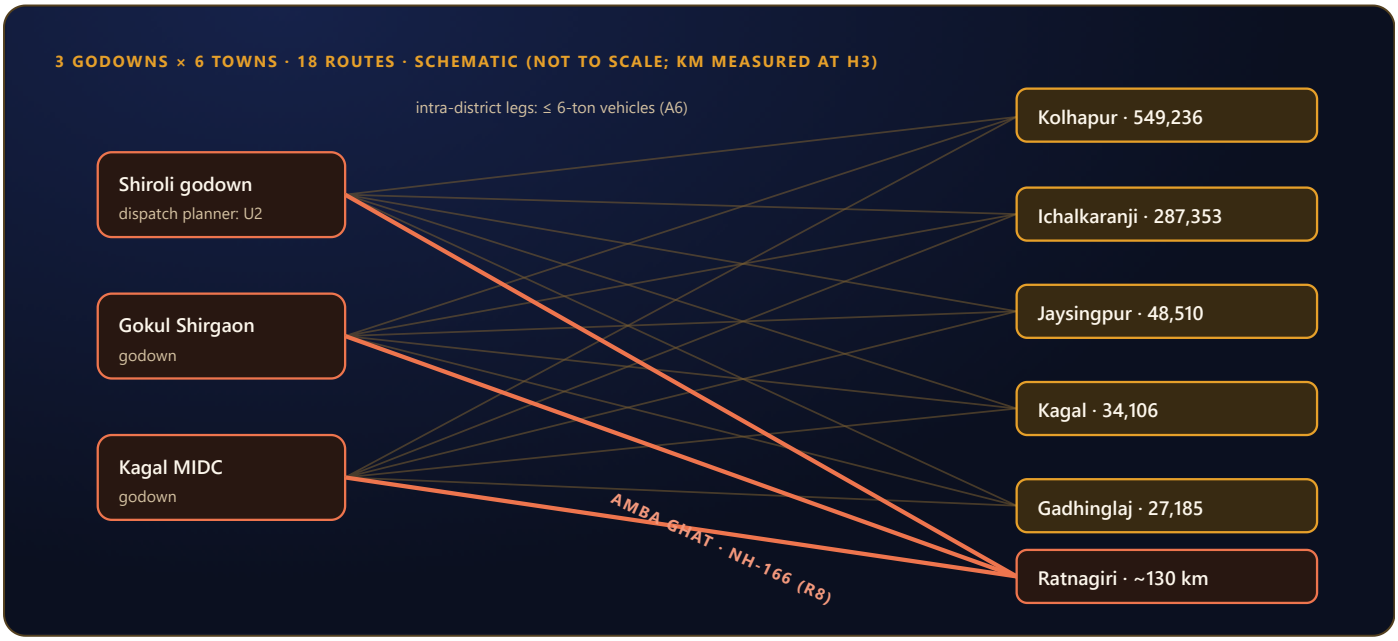


FIG 2 · *Census 2011 population drives each town's demand weight. Magenta = Ratnagiri legs, which normally run at the road-average rate and fall to 6-ton rates when Amba Ghat closes to heavy vehicles.*

04 Core User Journeys

Three views: **Plan & Compare**, **Proof** and **Disruption Lab**. The UI only formats numbers returned by the API.

Journey map

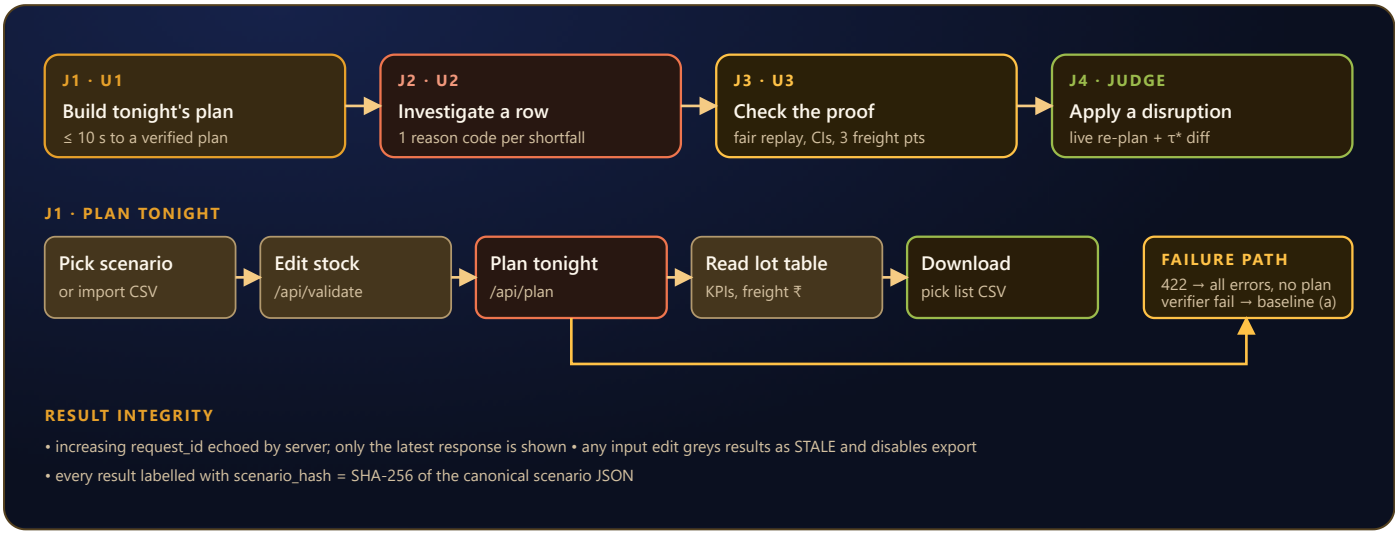


FIG 3 · *Four journeys, one data path. Failure never produces a silent plan.*

J1: U1 builds tonight's plan

#	Action	System response	Req
1	Selects a scenario and day, or imports a CSV export	Scenario loaded; batch forecasts read	FR-P0-01, -03
2	Edits godown stock	`POST /api/validate` returns errors and warnings	FR-P0-02
3	Clicks **Plan tonight**	`POST /api/plan`: eligibility → two LP solves → verifier → metrics → reason codes; rule-based baseline (a) alongside	FR-P0-04..08, -10
4	Reads lot-level table and KPIs; downloads the pick list	Godown → town → product → lot → units, freight ₹, life on arrival; CSV per godown	FR-P0-12

Done when: a verified plan is shown within 10 s (KPI-8). **Failure:** 422 lists all errors and no plan is shown. If the optimizer fails or its plan fails the verifier, baseline (a) is shown, labelled.

J2: U2 investigates a row

- **Allocation row:** lot, life at dispatch and on arrival, freight per unit, the route's τ*, and the segment filled.
- **Shortfall row:** exactly one reason code with facts (§7.6) — e.g. the towns that took the lot and the net ₹ value, after their own freight, of each one's last unit.
- **Done when:** every shortfall row in the demo scenarios shows one code whose facts match the plan (KPI-7).

J3: U3 checks the proof

#	Action	System response	Req
1	Opens **Proof**	Cached fair replay over 2 held-out 28-day windows; total ₹ bill per policy with 95% CIs	FR-P0-09
2	Picks a freight point (₹1.51 / ₹3.78 / ₹11.03)	Cached result at that point	FR-P0-09

#	Action	System response	Req
3	Opens **Is the headline real?**	The same replay scored two ways: lost units only vs full cost (the M3 trap)	FR-P0-11

J4: Judge Mode — apply a disruption LIVE DEMO

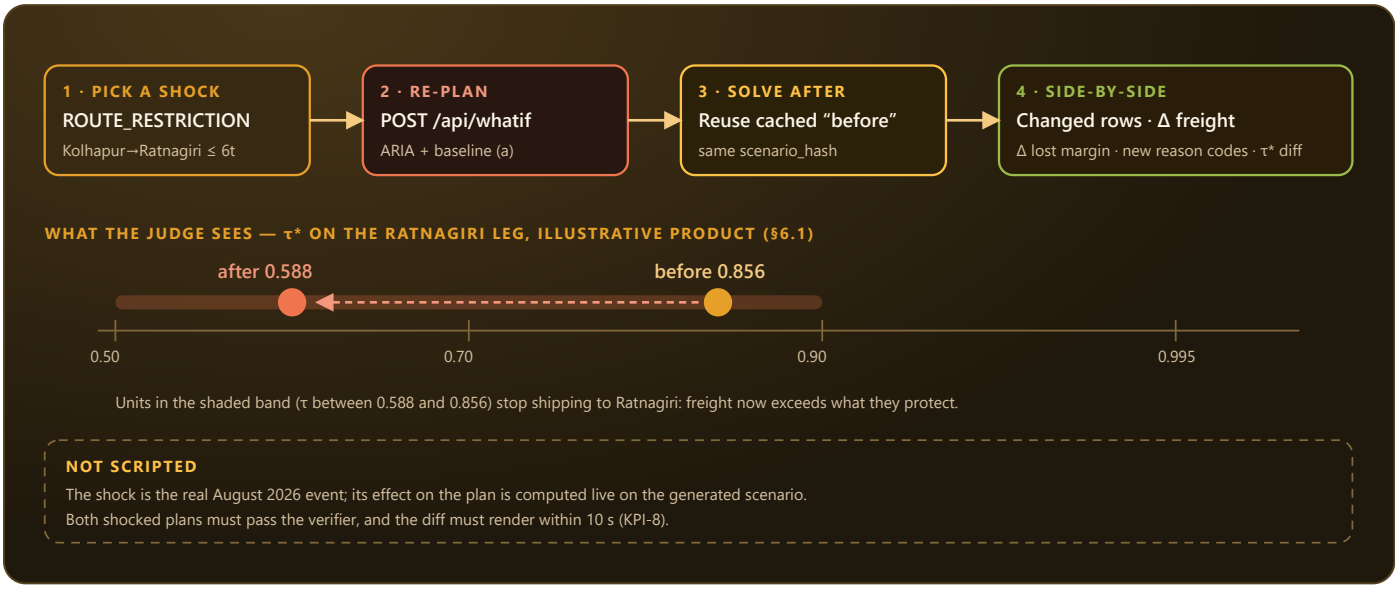


FIG 4 · *Judge Mode. Shocks:* `ROUTE_RESTRICTION` *(6-ton cap, as in Amba Ghat* R8*),* `ROUTE_CLOSURE`*,* `INVENTORY_LOSS`*. The §6.1 figures are for one illustrative product; real km are measured at H3.*

05 Functional Requirements

Owners: A optimizer & explanations · B data & forecasting · C baselines, verifier & replay · D validation, API, UI & demo.
Gates: early warning at **H13**; P0 gate at **H15**; no new P1 after **H17** (§8).

Requirement dependency map

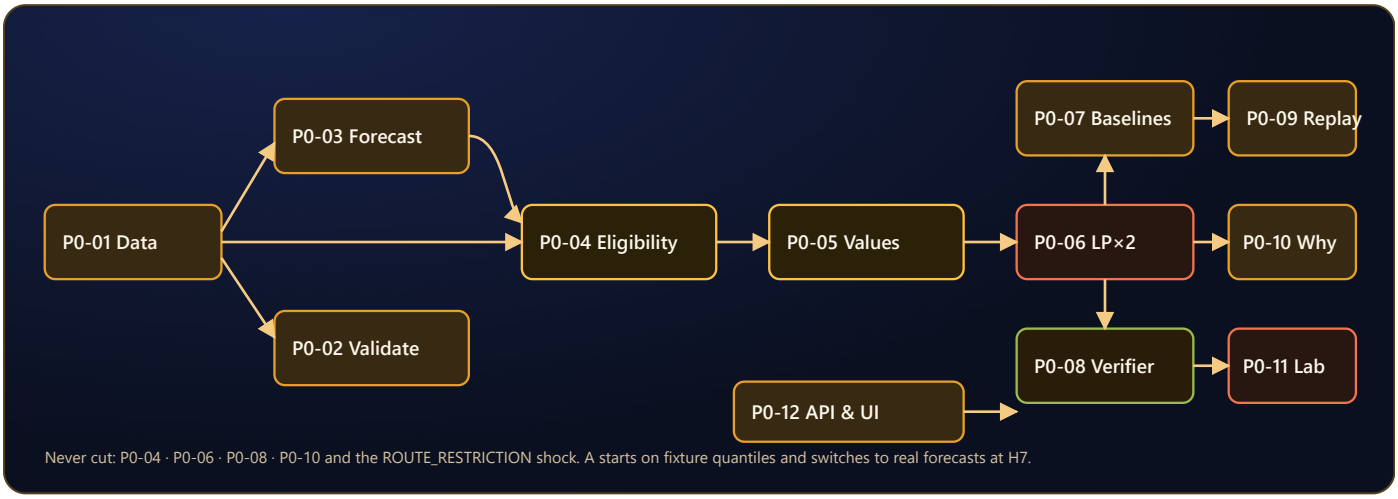


FIG 5 · *Build order: A builds 04/05/06 on fixture quantiles; P0-10 needs 06 + 08; P0-11 needs 06, 07 and 12.*

5.1 P0 — Must have (12 requirements)

ID	Requirement	Acceptance (testable)	h	Own
P0-01	**Data, scenarios & CSV import**	`make data` runs offline from USB; a fixture CSV runs forecast → plan → replay with no code change; every Appendix D parameter has a source and status	4	B
P0-02	**Validation:** Pydantic v2 + cross-record rules (App. A)	One fixture per rule returns the expected code and JSON path; errors block solving	1.5	D
P0-03	**Calibrated quantile forecasts**	9 non-crossing quantiles per (town, product, day); coverage table (nominal / raw / conformal) stored	3	B
P0-04	**Eligibility:** lot l may serve town s only if (i) an active route exists; (ii) `remaining_life(l) − transit_days ≥ min_days_at_receipt(k)` ; (iii) qty > 0 and life > 0	Boundary tests: equality eligible, one day short not; verifier V3 on every plan	1	A
P0-05	**Freight-aware segment values**	Unit test: 1 godown, 1 town, f = 0, ample stock gives the newsvendor quantile at $c_u/(c_u+h)$ R12, within one segment	1.5	A
P0-06	**Optimizer:** two network-flow LPs with HiGHS R15. Stage 1 maximises V; Stage 2 keeps only zero-reduced-cost arcs and prefers shortest-life lots, losing no V. Duals returned	Both OPTIMAL on bundled and 200 random scenarios; integral (V1); exact integer check V(stage 2) = V*; a FEFO fixture where stage 2 differs from stage 1	3.5	A
P0-07	**Baselines** (a), b′, b″, (c)	(a) identical over 10 runs and matches a hand-computed 2×2 fixture; b′ and b″ pass the verifier	3	C
P0-08	**Verifier, metrics, property tests:** V1–V8 separate from the solver; failing plan withheld & counted; one metrics function; P1–P7 on ≥ 200 random scenarios	One mutation test per V-code; metrics fixture exact to the paisa; suite green in < 60 s	3.5	C

ID	Requirement	Acceptance (testable)	h	Own
P0-09	**Fair replay**	Conservation: opening + received = sold + written off + closing, zero tolerance; no leakage (`train.max(day) < test.min(day)`); CIs written to `replay_results`	3.5	C
P0-10	**Reason codes & explain:** exactly one code + sub-code per shortfall row; explain card shows segments, τ*, freight/unit, life on arrival, indicative lot shadow price, parameter sources	`UNCLASSIFIED` = 0 for optimal plans on bundled and random scenarios (P6); every primal fact equals the plan or scenario value	2.5	A
P0-11	**Disruption Lab (Judge Mode)**	`INVENTORY_LOSS 0` gives an empty diff; `ROUTE_CLOSURE` on a used lane removes its rows; property P4 on random scenarios	4	D+A
P0-12	**API & UI:** FastAPI + Streamlit views, request integrity, CSV pick list, `demo.md`	J1–J4 complete with Wi-Fi off; every displayed KPI equals the API value (contract test)	5	D

Effort: 36 person-hours (A 9.5, B 7, C 10, D 9.5) against 60 productive hours (4 × 18 h, minus ~3 h sleep each).

FR-P0-01 — data, scenarios and CSV import

- **Demand:** M5 FOODS_3, top 100 products by volume R10. Each town gets one M5 store's demand shape (6 of 10 stores, in `mapping.yaml`), rescaled to the town's Census 2011 population share R16, keeping M5's own event calendar so features and outcomes stay aligned (Maharashtra festival dates only in the P1 festival shock).
- **CSV import:** a distributor's export with columns date, item, outlet/town, qty, price maps into the same schema and runs through the same batch pipeline (`POST /api/import` starts the batch forecast job, timed at H7; not on the 10 s interactive path).
- **Lots:** synthetic shelf-life classes within USDA FoodKeeper ranges R18. **Town stock:** on-hand units per town (lot, qty, remaining life); they fill the lowest demand segments before any shipment.
- **Network:** 3 godowns × 6 towns; road km precomputed once from OpenStreetMap R19.
- **Scenarios:** `normal_week` , `monsoon_squeeze` (scarcity, near-expiry lots at far godowns), `edge_cases` . `edge_cases` triggers codes 1–3, `monsoon_squeeze` triggers 3 and 4 (including SUPPLY_EXHAUSTED); no scenario may produce UNCLASSIFIED.

FR-P0-03 — calibrated quantile forecasts

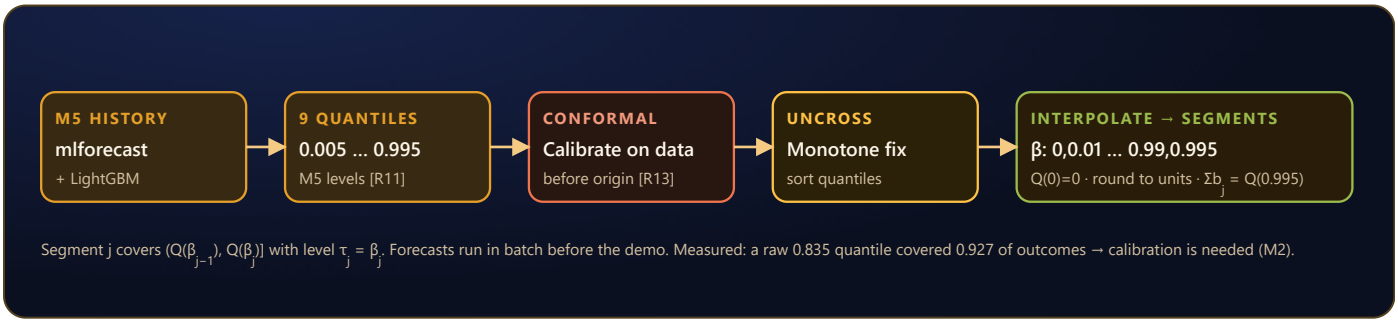


FIG 6 · *Forecast pipeline, from history to the integer demand segments the optimizer consumes.*

FR-P0-05 — freight-aware segment values

$$v_j = c_u \cdot (1 - \tau_j) - h \cdot \tau_j \; - \; \text{f charged on the arc that delivers the unit}$$

- c_u = price × distributor margin (3.5% base, 5% sensitivity R6) × optional town priority (default 1).
- **h** = unit cost × repo rate / 365 R14 — a lower bound; warehousing R1 would raise h and lower τ* further.
- **f** = ₹/t-km for the route's vehicle class R1 × kg/unit / 1000 × road km; kg per unit = Open Food Facts India category median R18.

- **Write-off credit** for an at-risk unit: $E_k(1 - \tau_j)$, with E_k = unit cost × (1 − salvage share). A near-expiry unit avoids its write-off only if it sells, which happens with probability $1 - \tau_j$.
- τ^* is therefore per (route, product). All values are integer paise; after rounding, adjacent low-price segments may tie, so V8 and P3 require only that no unfilled segment has a *strictly* higher value than a filled one.

FR-P0-07 — the baseline ladder

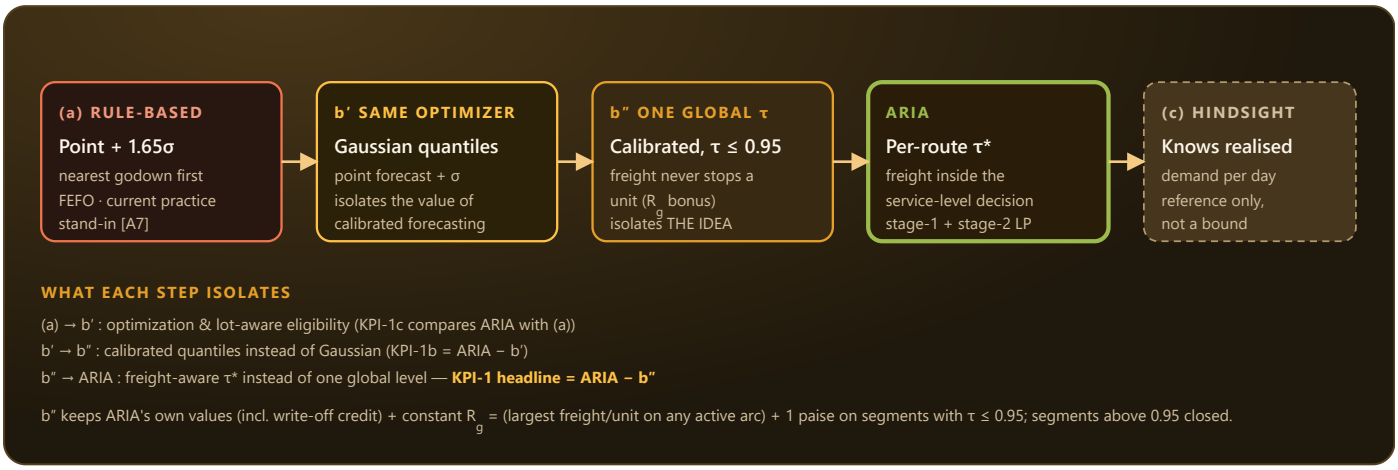


FIG 7 · *Each baseline changes exactly one thing, so each KPI names its own cause.*

FR-P0-09 — fair replay

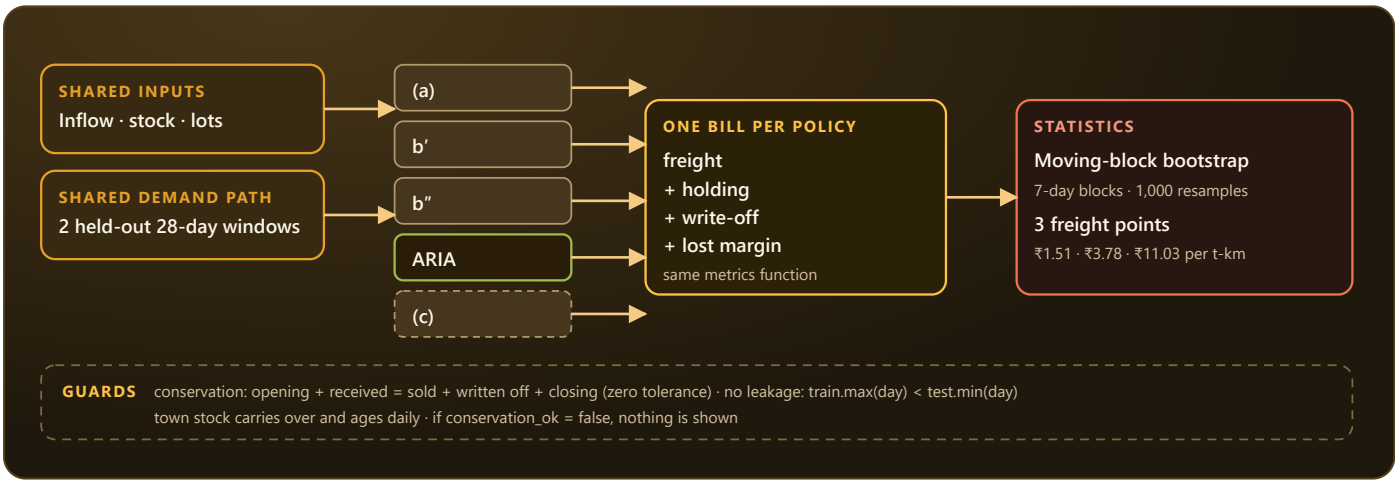


FIG 8 · *Why the M3 trap cannot happen: every policy pays the same bill from the same starting point.*

FR-P0-11 — Disruption Lab

- **Shocks:** `INVENTORY_LOSS {godown, product|*, pct}` · `ROUTE_CLOSURE {godown, town}` · `ROUTE_RESTRICTION {godown, town, max_vehicle:"6t"}` (re-prices the arc at max(current rate, NCAER 6-ton rate)).
- **Re-plan:** ARIA and (a) re-solved, server-side diff, τ^* change per (route, product). The "Is the headline real?" panel reads cached replay results; if no policy over-stocks on real M5, it says so.

Never cut: FR-P0-04, -06, -08, -10 and the `ROUTE_RESTRICTION` shock.

5.2 P1 — Should have (after a green H15 gate)

ID	Requirement	Acceptance	h
P1-01	Scarcity Value Map: rank godown × product by largest lot shadow price (screen only — duals non-unique); confirm top 5 by re-solving with +N units	Each confirmed probe has ΔV ≥ 0	2

ID	Requirement	Acceptance	h
P1-02	`FESTIVAL_SPIKE` shock: uplift learned from M5 event days, placed on a Maharashtra festival date R17 (labelled)	Diff renders; verifier passes	1.5
P1-03	Stockout masking; flag level from {0.01, 0.05, 0.10} by F1 against FreshRetailNet's true stock status R20	Bias on flagged days lower with mask	2.5
P1-04	Replenishment order per godown from the CFA (newsvendor on aggregated quantiles)	Order = critical-ratio quantile	2
P1-05	Ablations: flat freight; Gaussian vs conformal; no FEFO tie-break	KPI-1 CI for each	2
P1-06	Kolhapur mandi prices from data.gov.in API R21	≥ 1 Kolhapur market cached	1.5

5.3 P2 — Future

- Pilot on one Kolhapur distributor's export, with consent
- Multi-period planning with lead times
- Truckload constraints and vehicle routing (VRP)
- LLM that only rewords computed reason facts, checked so every number matches
- Marathi UI
- Minimum-fill floors per town

5.4 Out of scope

- User accounts
- A database server (embedded DuckDB + Parquet only)
- Cloud data
- Vehicle routing (VRP)
- Any LLM-generated number
- Automatic order execution — plans are advisory
- Any claim that M5 demand equals Kolhapur's

06 Innovation & Differentiation

6.1 The one idea: freight sets the service level

A unit filling quantile segment τ of a town's demand is worth shipping only if

$$c_u(1-\tau) - h\cdot\tau - f > 0 \iff \tau < \tau^* = (c_u - f) / (c_u + h)$$

Without freight this is the textbook newsvendor ratio R12. With freight, every (route, product) pair gets its own service level.

Accounting consistency. The plan and the replay charge holding the same way: h per unit-night on every shipped unit left unsold at a town (A9). Godown stock carries no holding charge in any policy, so the net value v(τ) − f above is exactly the expected change in the replay's one bill from shipping one more unit into that segment; realised demand is the only difference.

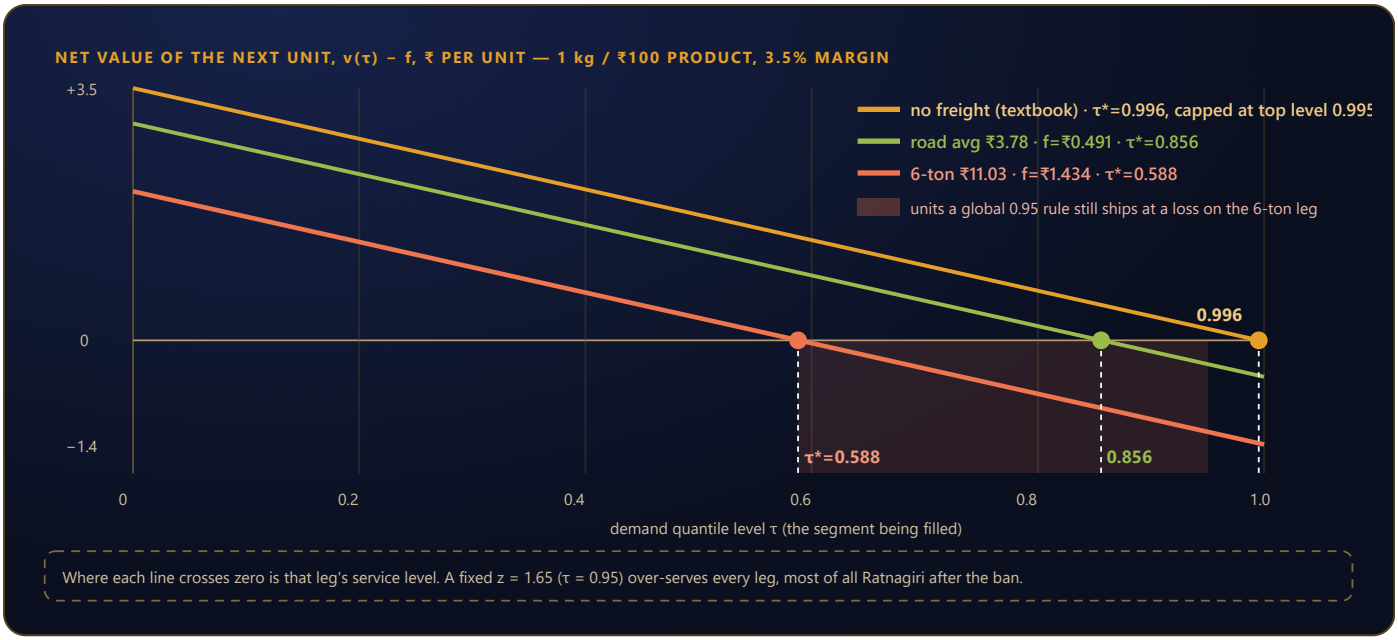


FIG 9 · *The whole idea in one picture. Freight shifts the line down; the zero-crossing — τ* — slides left.*

Worked example (one illustrative product). 1 kg unit per ₹100 of price; h from the repo rate R14; freight from R1. Distances are approximate road km, measured at H3.

Leg	Vehicle class	f per unit	τ* at 3.5% margin R6	τ* at 5% margin
No freight (textbook)	none	₹0	0.996*	0.997*
Kolhapur → Ichalkaranji, ~30 km	6-ton (A6)	₹0.331	0.902	0.931
Kolhapur → Ratnagiri, ~130 km	road average ₹3.78 (A6)	₹0.491	0.856	0.899
Kolhapur → Ratnagiri, Amba Ghat ban R8	6-ton ₹11.03	₹1.434	**0.588**	**0.711**

*Above the top M5 level of 0.995, so capped and flagged.

At distributor margins, **distance and vehicle class — not holding cost —** move the service level. Baseline b″ is that fixed rule inside the same optimizer, so KPI-1 measures this idea alone.

6.2 Why this is not a dashboard

Typical build A7	ARIA DESIGN
Point forecast + fixed buffer, or LLM text as "recommendation"	Calibrated quantiles; a lot-level plan from an optimizer with cited costs
Scarcity resolved by processing order	Scarcity resolved by net ₹ value per unit, with a provable "no lower-value recipient" property (P3)
Expiry checked at the godown	Shelf life checked **on arrival**; FEFO from a value-preserving tie-break
LLM-written explanations	Computed reason codes, provably complete for optimal plans (P6); every number cross-checked
No validity check	Independent verifier on every plan; a failing plan is withheld
"Stockouts down X%"	One full-cost bill on realised demand vs 3 named baselines, with CIs, and the "Is the headline real?" exposé (M3)

6.3 Comparison

Attribute	Blue Yonder A&R R24	Bizom / FieldAssist R25	Typical public build R26	ARIA
Godown → town allocation with freight	Yes	Not in public product pages (checked)	Flat ₹/km	**NCAER ₹/t-km by vehicle class**
Service level	Configurable	Not in public pages	Fixed z	**Per route, from margin, holding and freight**
Shelf life on arrival	Expiry-aware	Not in public pages	No	**Yes, lot level**
Proof	Vendor case studies	Vendor claims	None	**Fair replay; property tests**
Runs offline on one laptop	No	No (SaaS)	Yes (script)	**Yes, with UI**

We claim only that ARIA is open, cited, offline, verified and fairly tested.

6.4 Reused

LightGBM · mlforecast conformal intervals R13 · HiGHS R15 · FastAPI · Streamlit

Built

Freight-aware segment values · lot eligibility · two-LP model · verifier · reason codes · Disruption Lab · fair replay · Kolhapur network

MOAT — HONEST

The algorithms are public. The moat is the verified, cited pipeline plus a replay that runs on a distributor's own CSV export (P0). It could be rebuilt in weeks.

07 High-Level Architecture & Flow

7.1 Technology stack

Layer	Choice	Rationale and alternative considered
UI	Streamlit	Python-only, so all 4 members can edit it. React + Vite considered and rejected: a second language and a build step DESIGN
API	FastAPI + Pydantic v2, Python 3.11.9	Typed validation and OpenAPI; 3.11 wheels installed cleanly M1
Forecasting	mlforecast 1.1.0 + LightGBM 4.7.0	M5 evidence R11; conformal intervals R13. Small fit 0.51 s (M1); full 100-product, 9-quantile fit timed at H7
Optimization	highspy 1.15.1 (HiGHS simplex)	Duals; a 20,400 × 530,000 LP solved in 3.3 s (3.81 s with build) M1; our model is ~65k columns. OR-Tools SCIP considered; not needed once both stages are LPs
Data	Parquet + embedded DuckDB; scenario JSON	No server; offline
Hosting	One laptop, local; screen recording at H17 as fallback	Venue Wi-Fi measured at ~25 kB/s MEASURED

Versions pinned at H0. Wheels and datasets come from USB, because the online install took 459 s (M1).

7.2 System architecture

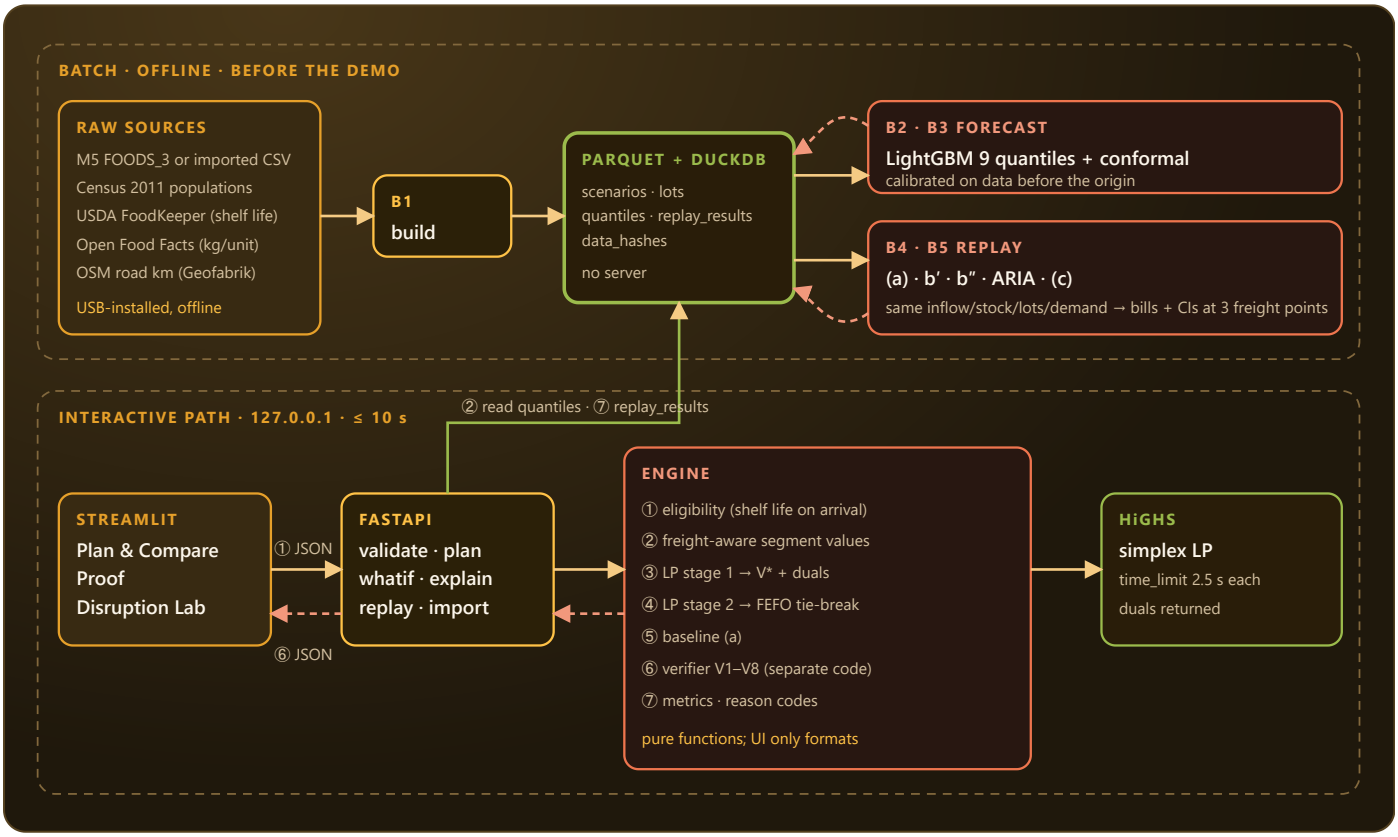


FIG 10 · *Architecture. Everything left of the dashed line of the demo runs before the demo; the interactive path never trains a model.*

7.3 /api/plan — step by step



FIG 11 · *Interactive path and latency allocation.*

7.4 Optimization model

Network

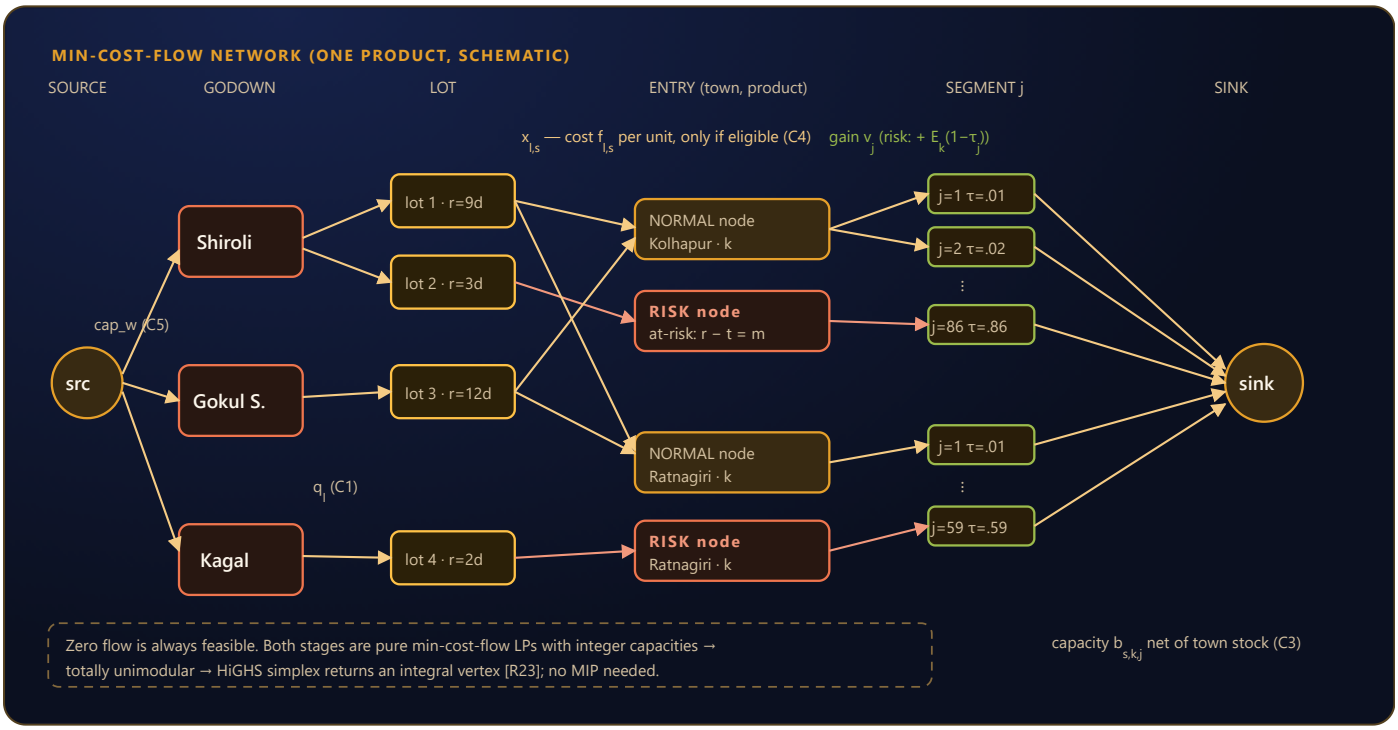


FIG 12 · *The allocation as a flow. Magenta paths carry at-risk lots and earn the write-off credit in proportion to their chance of selling.*

Variables

- $\mathbf{x_{l,s}}$ units shipped on eligible arc (l, s), freight $f_{l,s}$
- $\mathbf{y_{s,k,j}}$, $\mathbf{y^{risk}}$ units filling segment j from normal / at-risk lots
- $\mathbf{z_l = \Sigma_s\, x_{l,s}}$ flow through lot l

Constraints

- **C1** $z_l \le q_l$ · **C2** flow balance at entry nodes
- **C3** $y + y^{risk} \le b_{s,k,j}$ (net of town stock)
- **C4** arcs only for eligible (l, s) · **C5** $\Sigma\, z_l$ at w $\le cap_w$

Two stages, one value

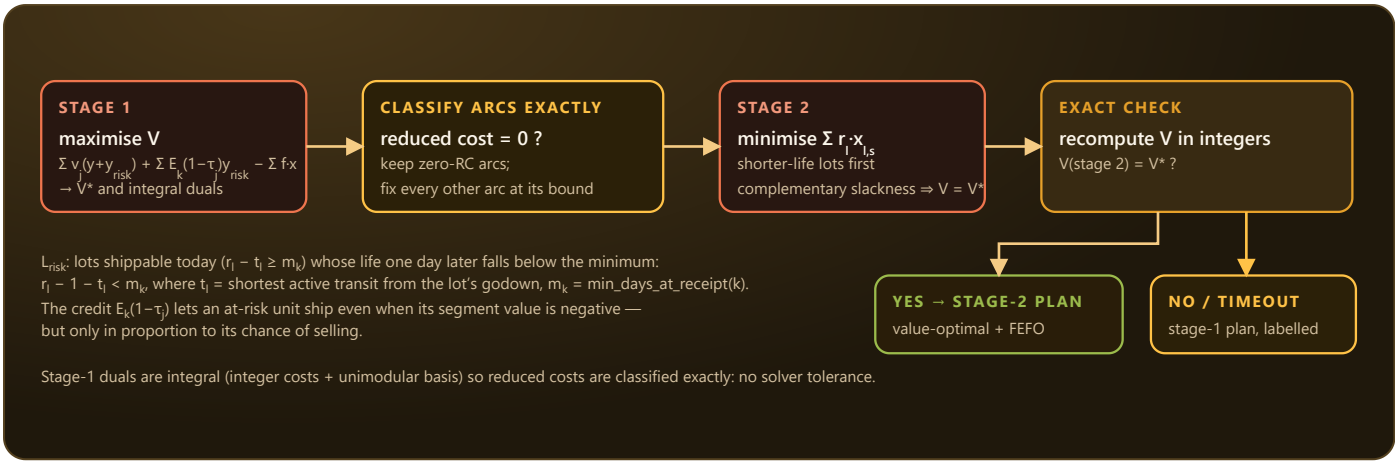


FIG 13 · *Exact FEFO tie-break that can never cost value.*

Properties (property-tested; both stages OPTIMAL)

ID	Property	Basis
P1	Every plan passes V1–V8	Verifier
P2	V*(ARIA) ≥ V(baseline a) **under the forecast** (expected value, not realised cost)	(a)'s shipments, mapped lowest-segment-first, are feasible for stage 1 — (a) is capped at the top modelled quantile and uses the same eligibility
P3	For any lot l there is no eligible town B whose next unserved segment value minus $f_{l,B}$ exceeds that of a unit l sent to town A	Otherwise moving that unit raises V*. Net values include the risk credit; holds with ties
P4	INVENTORY_LOSS and ROUTE_CLOSURE never increase V*; ROUTE_RESTRICTION never lowers freight so never increases V*; adding stock never decreases V*	Feasible set shrinks or costs rise. FESTIVAL_SPIKE (P1) not covered
P5	Relabelling every ID leaves V* unchanged	Order independence
P6	Exactly one reason code per shortfall row; UNCLASSIFIED = 0	Codes partition the cases (§7.6)
P7	Both plans are integral	Total unimodularity

> **NOT CLAIMED**
>
> That ARIA ships more units than (a), or that it wins on realised cost. The second is tested by KPI-1, not assumed. Model size ≈ 100 products × 6 towns × 100 segments (doubled where at-risk stock exists) + ≈ 5,400 shipment arcs: 65k–125k columns, under the `E_SIZE` guard.

7.5 Data model

Money is stored in integer **paise**; quantities in integer **units**.

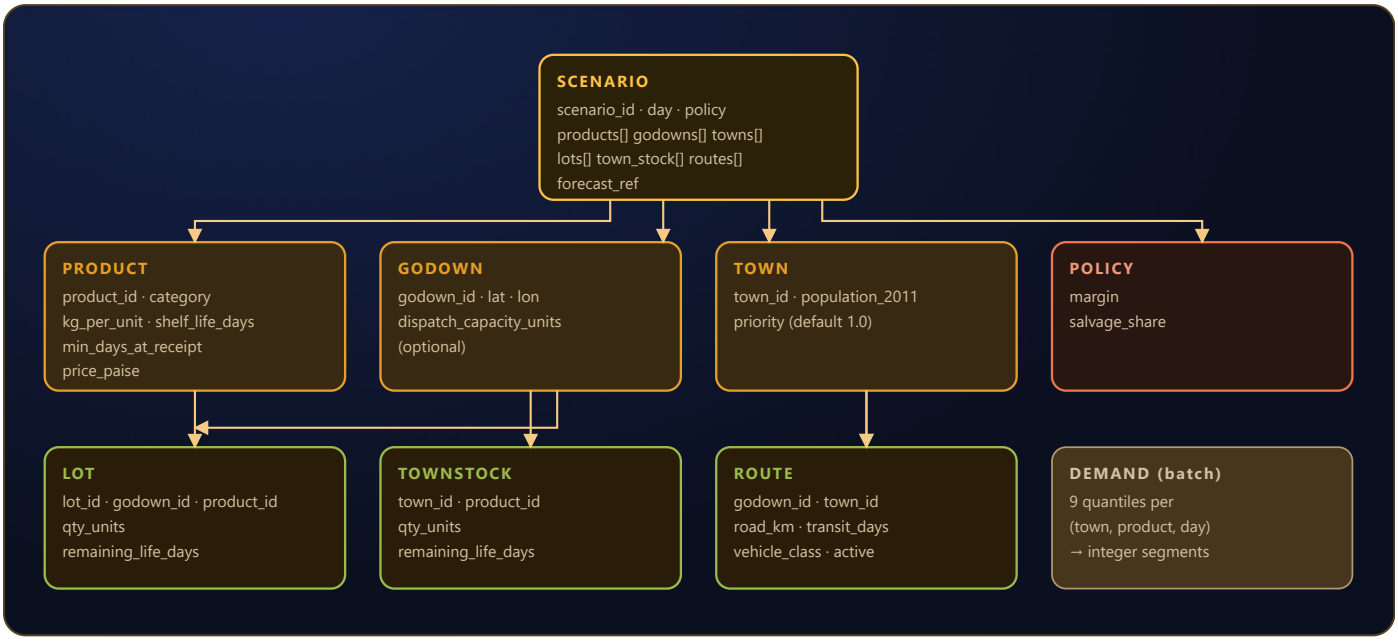


FIG 14 · *Entities (lots, town stock and routes reference products, godowns and towns by ID — checked by* `E_BAD_REF`*).*

7.6 Reason codes — one per shortfall row, first match wins

A **shortfall row** is a (town, product) whose on-hand plus shipped units are below its target T: T = Q(τ*) on its cheapest eligible route, or the median Q(0.5) when no eligible route exists (codes 1–2).

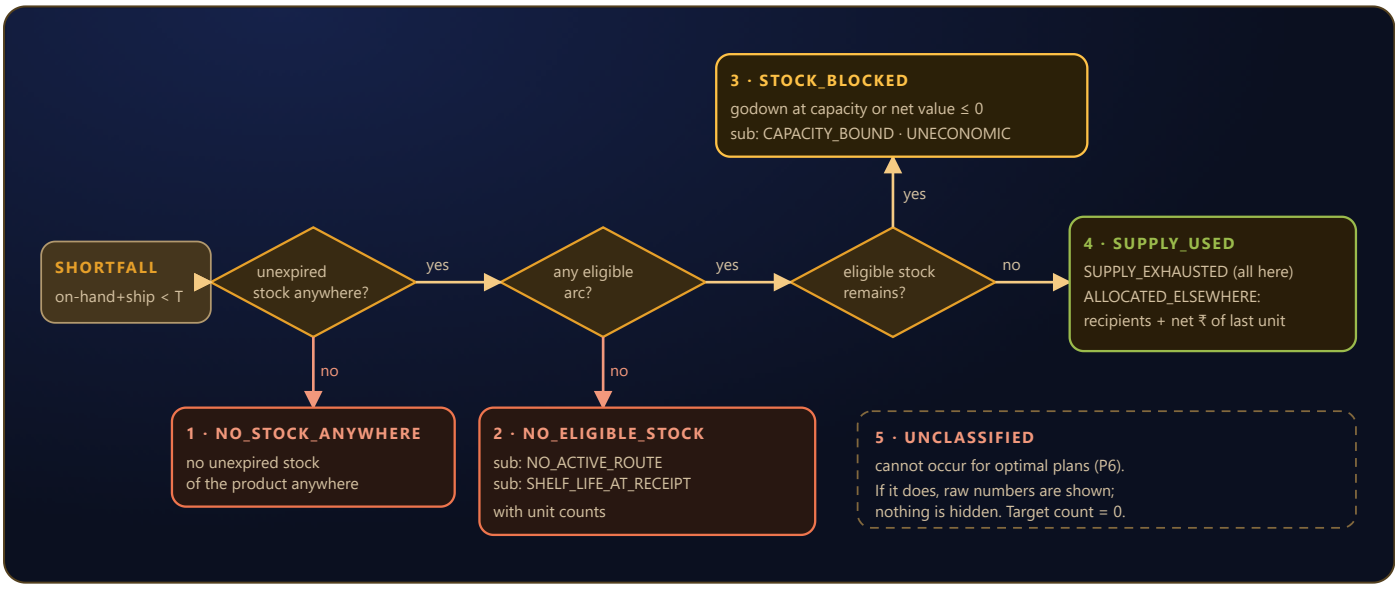


FIG 15 · *The codes partition every case, so an optimal plan yields exactly one. In baseline (a), lower-value recipients are flagged.*

7.7 API contracts

JSON bodies; money in integer paise. Requests that compute something carry `request_id`; responses echo it.

Method	Path	Request → Response	Errors
GET	`/api/health`	→ `{status, solver_version, data_hashes}`	—
GET	`/api/scenarios`, `/{id}`	→ list · `Scenario`	404

Method	Path	Request → Response	Errors
POST	`/api/import`	CSV file → 202 `{scenario_id, rows, warnings[], forecast_job_id}`	422 missing columns
POST	`/api/validate`	`{request_id, scenario}` → `{valid, errors[], warnings[], scenario_hash}`	400
POST	`/api/plan`	`{request_id, scenario}` → `{request_id, scenario_hash, solve_ms, recommended, aria: Plan, baseline_a: Plan}`	422, 500
POST	`/api/whatif`	`{request_id, scenario, shocks[]}` → `{request_id, before, after, diff:{aria, baseline_a}, tau_changes[]}`	422 (shock index)
GET	`/api/explain/{plan_id}?town=&product=`	→ `{segments[], tau_star, freight_per_unit_paise, life_on_arrival, lot_shadow_price_paise, reason, params:[{name,value,source}]}`	404
GET	`/api/replay/{id}`	→ `{per_policy:{bill_paise, freight, holding, writeoff, lost_margin, fill_rate}, kpi1:{diff_paise, ci_low, ci_high}, freight_point, conservation_ok}`	404; nothing shown if `conservation_ok=false`

Plan = `{stage1_status, stage2_status, stage1_only, allocations[], shortfalls[], metrics, verifier:{passed, violations[]}}` . `aria` is null if the optimizer has no solution or its plan fails the verifier; baseline (a) is then shown, labelled. All endpoints bind to 127.0.0.1 with no authentication (single-user local demo); client timeout is 15 s, above the 10 s server budget.

7.8 Failure behaviour

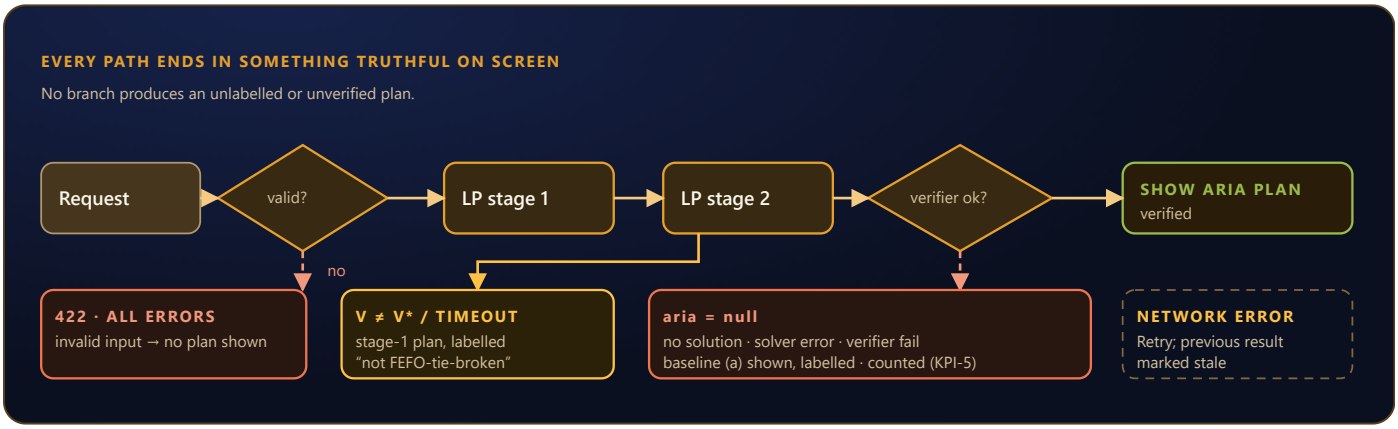


FIG 16 · *Failure behaviour state diagram.*

08 Feasibility & 18-Hour Build Plan

The build runs 18 hours (H0 = build start, H18 = end). Each member sleeps about 3 h in rotation, inside off-peak blocks. Riskiest pieces first: offline kit, then model and verifier.

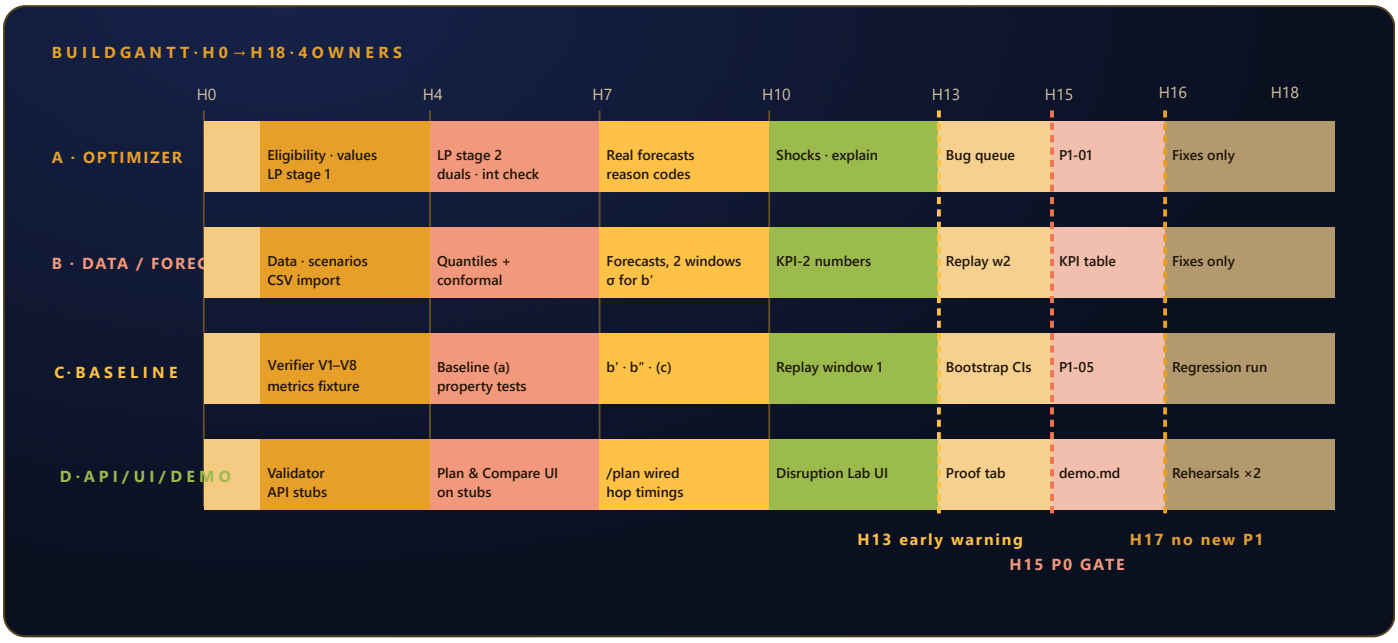


FIG 17 · *Build plan with gates. 1-hour blocks (H0–1, H1) are formulation, USB kit, repo and shells.*

Exit checks

- H0–1 Wi-Fi not needed
- H1–4 unit tests green
- H4–7 stage 1 OPTIMAL on `normal_week`
- H7–10 J1 + J2 end to end
- H13 warning · H15 P0 gate · H17 P1 freeze

Cut order if H15 is red

1. Replay window 2
2. The exposé panel (keep the KPI-1 card)
3. Freight points down to ₹3.78 and ₹11.03

P0-04, -06, -08, -10 and ROUTE_RESTRICTION are never cut.

Risks

Risk	Likelihood	Impact	Mitigation	Fallback
Install or downloads fail (459 s online; ~25 kB/s Wi-Fi)	High	High	Python 3.11 wheelhouse + datasets on USB via phone hotspot	One machine runs the backend
"Why US data?"	High	Medium	No open Indian daily sales data found [R27]; Indian costs, roads, populations, festivals; CSV import in P0	Present results as method evidence
Formulation bug	Medium	High	Verifier; P1–P7; hand fixtures	Withhold the plan; show (a)
Replay credits over-stocking (M3)	Seen in spike	High	One bill; (a), b′, b″; conservation test	Report honestly
Full forecast fit slow	Medium	Medium	Timed at H7; top-50 products as labelled fallback	Smaller set
KPI-1 fails	Unknown	Medium	Pre-registered response (§9)	Ablations

09 Success Metrics

PRE-REGISTERED

All pass rules are fixed now, before any replay; results are filled in at H13–17. No KPI is a guessed percentage: each compares against a named baseline (FR-P0-07), a provable property, or a cited limit. "CI" = 95% moving-block bootstrap CI (7-day blocks, 1,000 resamples). Sample sizes (≥ 200 scenarios, 30 calls, 2 windows) are design choices.

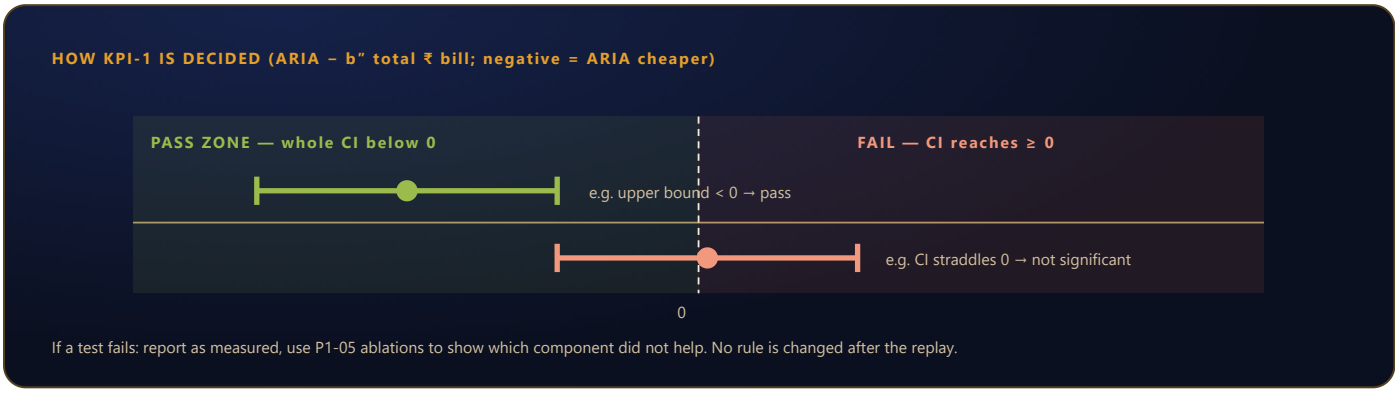


FIG 18 · *Schematic only — illustrative intervals, not results.*

ID	Metric	Pass rule	Method	Result
KPI-1 HEADLINE	Total ₹ bill on realised demand at each route's own vehicle-class rate (A6): ARIA − b″	**Upper CI bound < 0**	Replay over 2 windows × 28 days × 6 towns × 100 products	To fill
KPI-1b	Same, ARIA − b′ (forecast contribution)	Upper CI bound < 0	Replay	To fill
KPI-1c	Same, ARIA − (a) (vs rule-based stand-in)	Upper CI bound < 0	Replay	To fill
KPI-2	Scaled pinball loss over 9 quantiles, sales-weighted as in M5 [R11], ARIA ÷ seasonal-naive	Upper CI bound < 1.0; conformal coverage closer to nominal than raw at every level	utilsforecast; moving-block bootstrap	To fill
KPI-3	(freight + holding + write-off) per unit sold, ARIA − (a)	Upper CI bound ≤ 0 — fill rate not bought with cost (M3 guard)	Replay	To fill
KPI-4	Write-off units, ARIA − (a) (the brief's "wastage")	Upper CI bound ≤ 0	Replay	To fill
KPI-5	Plans passing V1–V8 / all solver outputs, counted before withholding	100%; withheld = 0	Harness & demo runs	To fill
KPI-6	Properties P1–P7 on ≥ 200 random scenarios	100%	pytest	To fill
KPI-7	Reason-code completeness and fact fidelity, optimal plans	UNCLASSIFIED = 0; 100% of primal facts match	pytest	To fill
KPI-8	`/plan` and `/whatif`, click → rendered	p95 and max ≤ 10 s [R9] over 30 calls (n too small for a CI, so max also reported)	Client-side timer	To fill
KPI-9	Outbound connections during the demo	0	`netstat`, Wi-Fi off	To fill
KPI-10	KPI-1 robustness	Rule holds with all routes at ₹3.78 and all at ₹11.03; ₹1.51 reported	Replay at 3 points	To fill

Display only (no pass rule): % of the (a)-to-(c) gap closed = $(\Sigma C_a - \Sigma C_p)/(\Sigma C_a - \Sigma C_c)$, computed as a ratio of sums; shown as "n/a" when the CI of $(\Sigma C_a - \Sigma C_c)$ includes 0. Whether (c) lands below every policy is a diagnostic, not a gate.

Demo hypotheses (reported whatever the outcome)

H1

On `normal_week`, ARIA and (a) differ mainly in freight and write-off.

H2

On `monsoon_squeeze`, ROUTE_RESTRICTION lowers Ratnagiri's τ* and shipped units while P2 still holds.

H3

The exposé panel shows a gap between lost-units-only and full-cost scoring for at least one policy.

After the hackathon (needs real users): time to produce a nightly plan; share of plans accepted unedited; real write-offs and freight ₹ against a control.

10 Data Sources & Assumptions

10.1 Measured by the team

M1–M3 come from a spike on 24 Oct 2026 (Windows CPU laptop, Python 3.11.9, synthetic Poisson demand): timings and traps, **not results**.

459 s
M1 · online install → USB kit

3.3 s
M1 · 530k-column LP (3.81 s with build); HiGHS 0.002 s vs CBC 0.31 s

0.927
M2 · a raw 0.835 quantile covered 92.7% → calibrate

5.6×
M3 · "86% fewer lost units" came from 5.6× more stock and 13% more freight

Venue Wi-Fi: ~25 kB/s during a download. Small mlforecast fit: 0.51 s.

10.2 Assumptions

ID	Assumption	Validation	Impact if wrong
A1	U1–U3 are composites	On-site conversation, H0–1	Details change; mechanism does not
A2	One M5 store's daily shape per town, scaled to Census population	Alternative mapping in P1	Absolute ₹ change; comparison stays fair
A3	Daily godown inflow = seasonal-naive network forecast, split equally, identical for every policy	Corner check: all inflow at Shiroli	Changes how often rationing happens
A4	Stores enforce a minimum remaining shelf life on receipt (synthetic values)	On-site conversation	Eligibility tighter or looser
A5	Unmet demand is a lost sale	Not testable in 18 h	Over-states lost margin
A6	Intra-district legs use ≤ 6-ton vehicles; Ratnagiri normally at road-average rate; linear freight per unit	Ask U2 about the truck mix	f shifts; covered by 3-point sweep
A7	Baseline (a) stands in for current practice	On-site conversation	Baseline label changes
A8	Lot shelf lives synthetic, within FoodKeeper ranges	None without real data	FEFO results illustrative

ID	Assumption	Validation	Impact if wrong
A9	M5 sell-out stands in for kirana order demand; town-level stock stays the distributor's until sold, so distributor bears h	CSV import of real orders (FR-P0-01)	Holding and lost-sale attribution shift

11 Limitations

Limitation	Effect	Mitigation
Demand shape from M5 (US); no open Indian daily sales data found [R27]	The pattern is not Kolhapur's	Indian scale, costs, roads, festivals; relative claims only; CSV import (P0); pilot (P2)
Synthetic lots and shelf lives	FEFO and write-off results illustrative	FoodKeeper ranges; labelled everywhere
h uses the repo rate (a lower bound)	τ* slightly overstated	Direction stated; warehousing cost as P1 sensitivity
Single period, two echelons, linear freight; no truckloads	Simpler than real dispatch	P2-02, P2-03
Value ordering can starve small towns	Equity not modelled	P2-06
Personas not interview-based	Pain points may be off	Interview at H0–1

A–D Appendices

Appendix A · Validation rules

Errors: `E_REQUIRED`, `E_TYPE`; `E_RANGE` (negative quantity or cost; remaining life above shelf life; minimum days above shelf life; priority outside 0.1–10; salvage outside 0–1; shock % outside 0–100); `E_DUPLICATE_ID`, `E_BAD_REF`; `E_SIZE` (> 300,000 model columns, derived from M1 and re-measured at H7).

Warnings: `W_NO_ELIGIBLE_SUPPLY`, `W_EXPIRED_STOCK`, `W_INACTIVE_ROUTE`.

Appendix B · Verifier (code separate from solver)

- **V1** integer, non-negative quantities
- **V2** lot stock respected
- **V3** every shipment on an eligible arc
- **V4** on-hand + shipped ≤ $Q_{top} = \Sigma_j b_j$
- **V5** dispatch capacity respected
- **V6** shortfall = T − (on-hand + shipped)
- **V7** all IDs exist
- **V8** segments within width; no unfilled segment strictly higher-valued than a filled one

Appendix C · Baseline (a), deterministic

```
for each product k (ascending id):
  for each town s by (descending point forecast, ascending id):
    need = min( round(point_forecast[s,k] + 1.65 * sigma[s,k]), Q_top[s,k] )   # capped at top modelled quantile
    for each godown w with an active route to s, by (ascending freight/unit, ascending id):
      for each lot l at w of k, by (ascending remaining life, ascending id):
        if eligible(l, s):                    # same FR-P0-04 rule
          q = min(need, lot_left[l], cap_left[w]);  ship q;  update need, lot_left, cap_left
```

Appendix D · Parameter registry

Parameter	Value	Status	Source
Demand quantiles	9 levels → 0.01 segments	Learned	R10, R11
Margin	3.5% base; 5% sensitivity	Cited	R6
Holding h	unit cost × 5.50% / 365 (lower bound)	Cited (secondary)	R14
Freight	₹1.51 / ₹3.78 / ₹11.03 per t-km by class	Cited	R1
kg per unit	OFF India category median	Learned	R18
Town weights	Census 2011: Kolhapur 549,236 (verified); Ichalkaranji 287,353; Jaysingpur 48,510; Kagal 34,106; Gadhinglaj 27,185 (cross-check H1); Ratnagiri fetched at H1	Cited	R16
Festival dates	Ganesh Chaturthi 14 Sep 2026; Laxmi Pujan 8 Nov 2026 (confirm against state order)	Cited	R17
Write-off credit	$E_k(1 - \tau_j)$, E_k = unit cost × (1 − salvage share); salvage swept 0–1	Derived	definitional
Shelf life; min days at receipt	Synthetic within FoodKeeper ranges	Synthetic (A4, A8)	R18
Global level for b″ and (a)	0.95 ≈ Φ(1.65)	Baseline definition	n/a
10 s ceiling	Attention limit	Cited	R9
`E_SIZE`	300,000 columns	Derived from M1	M1
Hop shares; HiGHS time limits; ≥ 200 scenarios; 30 calls; < 60 s suite; 1,000 resamples; 7-day blocks; 2 windows	As stated	Design choice (not sourced)	n/a

REF References (accessed 24 October 2026)

R1 NCAER for DPIIT, *Assessment of Logistics Cost in India 2023-24*, Sept 2025: road ₹3.78/t-km (pp. 21, 83); 6-ton ₹11.03 and 55-ton ₹1.51 (pp. 21, 88, §4.6.6); warehousing (p. 118, §4.10.3). ncaer.org/wp-content/uploads/2025/09/NCAER_Logistic_Study_Report_Sept2025.pdf

R2 ITLN, "Indian logistics costs at 7.97% of GDP", 30 Sep 2025 (secondary). itln.in/logistics/indian-logistics-costs-at-797-of-gdp-new-study-reveals-1356654

R3 Business Today, kirana stores and quick commerce, 19 Nov 2024. businesstoday.in/latest/economy/story/indian-kirana-stores-feel-the-heat-as-quick-commerce-sees-rapid-rise-454232-2024-11-19

R4 FMI, global out-of-stock study, 14 Oct 2002. fmi.org/newsroom/news-archive/view/2002/10/14/new-industry-study-explores-root-causes-of-global-out-of-stock-dilemma

R5 "Stockout", Wikipedia (secondary summary of Corsten & Gruen). en.wikipedia.org/wiki/Stockout

R6 Kirana Club, FMCG distributor and kirana margins. kirana.club/resources/fmcg-distributor-kirana-margins

R7 Deccan Herald, AICPDF on near-expiry products. deccanherald.com/business/companies/fmcg-firms-dumping-near-expiry-products-through-quick-commerce-distributors-fed-3263393

R8 Punekar News, "Amba Ghat closed to heavy vehicles amid landslide threat after heavy rain", 1 Aug 2026. punekarnews.in/?p=241185

R9 J. Nielsen, "Response Times: The 3 Important Limits", NN/g (1993, updated 2014). nngroup.com/articles/response-times-3-important-limits/

R10 M5 dataset, Zenodo (CC BY 4.0); M5 paper. zenodo.org/records/12636070 · arxiv.org/pdf/2102.13209

R11 M5 Accuracy and Uncertainty results, IJF 2022. ideas.repec.org/a/eee/intfor/v38y2022i4p1482-1491.html

R12 Newsvendor model (critical ratio). en.wikipedia.org/wiki/Newsvendor_model

R13 Nixtla mlforecast, prediction intervals (conformal). nixtlaverse.nixtla.io/mlforecast/docs/how-to-guides/prediction_intervals.html

R14 RBI MPC, 7 Oct 2026: repo rate 5.50% (secondary: GKToday; corplawupdates.in). gktoday.in/rbi-hikes-repo-rate-by-25-basis-points-to-5-50/

R15 HiGHS (MIT) and highspy. github.com/ERGO-Code/HiGHS

R16 Census of India 2011. censusindia.gov.in · citypopulation.de/en/india/maharashtra/

R17 Maharashtra public holidays 2026 (confirm against state order). maharashtra.gov.in

R18 USDA FoodKeeper (CC0); Open Food Facts (ODbL). catalog.data.gov/dataset/fsis-foodkeeper-data · world.openfoodfacts.org/data

R19 Geofabrik India Western Zone (OSM, ODbL). download.geofabrik.de/asia/india.html

R20 FreshRetailNet-50K (CC BY 4.0). huggingface.co/datasets/Dingdong-Inc/FreshRetailNet-50K

R21 data.gov.in, daily mandi prices API. data.gov.in

R22 S. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach*, 4th ed., Pearson.

R23 R. K. Ahuja, T. L. Magnanti and J. B. Orlin, *Network Flows: Theory, Algorithms, and Applications*, Prentice Hall, 1993.

R24 Blue Yonder, Allocation & Replenishment. blueyonder.com/solutions/retail-planning-and-category-management/allocation-and-replenishment

R25 Bizom (GetApp); FieldAssist (SoftwareSuggest). getapp.com/all-software/a/bizom/ · softwaresuggest.com/fieldassist

R26 ak6319/fmcg-distribution-optimization, GitHub (one of several public builds of this pattern).

R27 Team Cyberpunk, Indian open-data check (data.gov.in, Kaggle, Hugging Face, Agmarknet, HCES), 24 Oct 2026: `PS-selection/research/PS2_indian_data_check.md`

> **END OF DOCUMENT**
>
> ARIA · Team Cyberpunk · TRIWIZATHON 2026 · Problem Statement 2. **Nothing has been built yet**; the pass rules in §9 are fixed before any result exists.